

DAY 11 SUBQUERY

■ SQL Theory Notes – Subqueries, Joins, and Set Operations

◆ 1. Subqueries :

A subquery is a query inside another query, enclosed in parentheses.

It is used when we need a result from one query to use in another query.

Types of subqueries:

Single-row subquery → returns one value.

Multi-row subquery → returns multiple values.

Nested subquery → query inside another subquery

select * from customers

where creditLimit = (select max(creditLimit)from customers);

1. it give me the bill number.

select orderNumber,sum(total) from

(select orderNumber,

QuantityOrdered * priceEach as total

from orderdetails) as bill

group by orderNumber;

2.it give the output as the table.

select distinct

-- customerNumber,

-- customerName,country,

salesRepEmployeeNumber

from customers;

subquery always in the brackets.

select * from employees

where employeeNumber in

(select distinct

salesRepEmployeeNumber

from customers where country = 'USA');

```

use home;

create table cost (name varchar(255),costprice int);

insert into cost ( name ,costprice)

values('shirts',100),

('Jeans',200),

('Googles',300)

select * from cost;

```

CREATING OF THE TABLE FOR THE EXECUTION OF THE SUBQUERY

```

create table sales(name varchar(255),sellingPrice int);

insert into sales(name,sellingprice)

values('shirts',300),

('Jeans',800),

('Googles',1000)

```

1). Selecting the items using the subquery

```

select * from sales;

update sales

set name='T_shirts'

where sellingprice=1000;

```

-- it is the join

```

1) select * from cost c join sales s on c.name=s.name;

select name from cost union all select name from sales order by name;

2) select * from cost c right join sales s on c.name=s.name

union

select * from cost c left join sales s on c.name=s.name;

3) select * from customers c join employees e on c.salesRepEmployeeNumber=e.employeeNumber

4) select * from orders join customers on orders.customerNumber= customers.customerNumber

```

IMPORTANT POINTS

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-- pehle table ki foreign key and dusre table ki primary key use krenge.

-- one to one matching.we have done so far

-- next is one to many and many to one.

```

E) TYPES OF THE RELATION USED IN THE SUBQUERIES

One-to-One, One-to-Many, Many-to-One Relationships

One-to-One: Each row in table A relates to only one row in table B.

One-to-Many: One row in table A relates to multiple rows in table B (e.g., customers → orders).

Many-to-One: Multiple rows in table A relate to one row in table B.

Many-to-Many: Handled by a linking table (e.g., orderdetails links orders and products)